Low-Cost 6-Bit DAC



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For controlling the load of appliances such as cooling fans, low-wattage heaters, thermostats, low-wattage light sources, small electrical toys and test benches for loudspeakers, we need a power source whose voltage can be controlled in small steps and is capable of providing current of more than 1A. For that, we require low-resolution digital-to-analogue converters (DACs) with three to seven bits.

Here is the circuit capable of setting output voltage between 1.25V and 15V in 64 steps. The circuit can be adapted for a lot of applications.

Circuit and working

The circuit of the low-cost 6-bit DAC is shown in Fig. 1. The DAC is built around IC 7406, hex inverter (IC3). We may also use IC 7407 with six followers without changing the PCB. Steps are generated with the help of 6-bit digital input code D0 (LSB) through D5 (MSB) at CON2. Consequently, 64 combinations are possible starting from 000000 to 111111. At each combination, you will have a pre-determined output voltage between 1.25V and the possible maximum 15V.

Inputs D0 through D5 are TTL and CMOS compatible. These can be generated by microcontrollers, parallel-interface adapters such as PPI8255A, PIA6820/1 and Z80-PIO. In the simplest case, inputs can be driven with switches connecting inputs D0 through D5 to ground 0V or to 5V.

The size of the steps is programmable with trimmer potentiometers VR1 through VR6. Consequently, we can produce regular or irregular steps according to the need, depending on the characteristics of the load being controlled.

You can set any output voltage with any potentiometer between 1.25V and the maximum. For example, if you have a transformer for 18V AC, you can set outputs between 1.25V and around 15V with any potentiometer.

For adjustment in the simplest case, apply a set of seven test codes, as listed below, on CON2; output on CON3 will be as under.

Test code with 7406 (invert with 7407):

000000 Vout = Vmax (unadjustable)
000001 Vout = 1.25V (VR1)
000010 Vout = 3V (VR2)
000100 Vout = 5V (VR3)
001000 Vout = 7.5V (VR4)
010000 Vout = 9V (VR5)
100000 Vout = 12V (VR6)

The maximum output voltage on CON3 and CON4 is with code 111111 on the outputs of IC3 and depends on the input voltage of IC2. Please note that, if you use 7407, the codes will be non-inverted, and if you use 7406, the codes will be inverted.

The 6-bit input digital code D0

PARTS LIST

Semiconductors:
IC1 - 7805, 5V regulator
IC2 - LM317, positive-adjustable regulator
IC3 - 7406, hex inverter buffer (or 7407 non-inverting buffer)
BR1 - 1A bridge rectifier module
D1-D4 - 1N4007 rectifier diode
LED1 - 5mm red LED

Resistors (all 1/4-watt, ±5% carbon, unless stated otherwise):
R1 - 100-ohm, 2-watt
R2 - 120-ohm
R3 - 680-ohm
R4 - 2.2-kilo-ohm
VR1-VR6 - 1.5-kilo-ohm potentiometer

Capacitors:
C1 - 0.1µF, 400V polyster
C2 - 1000µF, 25V electrolytic
C3, C4, C6 - 0.33µF ceramic
C5 - 22µF, 16V electrolytic
C7 - 1µF, 25V electrolytic

Miscellaneous:
CON1, CON3-CON5 - 2-pin connector
CON2 - 7-pin connector
X1 - 230V AC primary to 18V, 1A secondary transformer
F1 - Fuse, 500mA

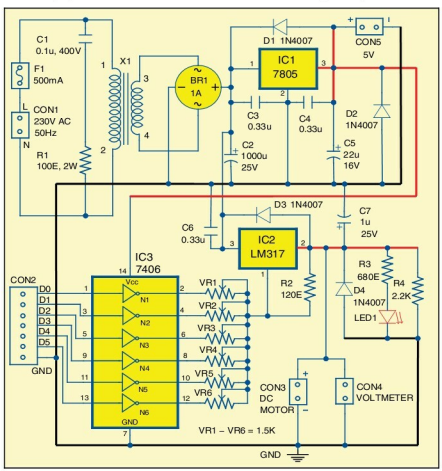


Fig. 1: Circuit of the low-cost 6-bit DAC